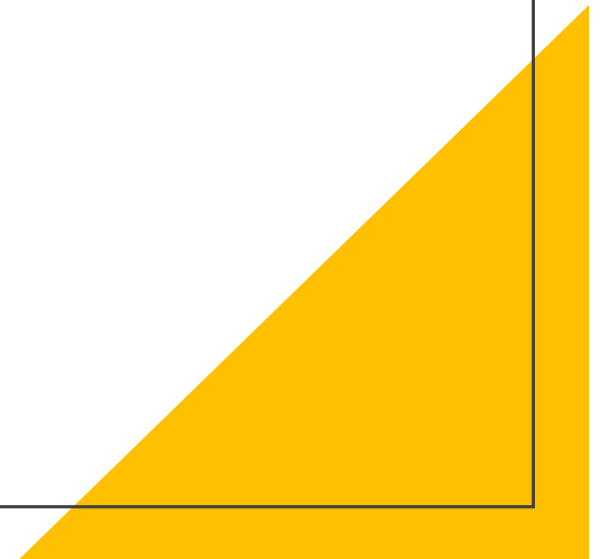


CSE 445

Lecture 8.1

Logistic Regression



Linear model for cond. Probability estimation

For data $x \in \mathbb{R}^d$, classify and return probabilities using a linear function

$$w_1x_1 + w_2x_2 + \cdots + w_dx_d + b = w \cdot x + b$$

where $w = (w_1, \dots, w_d)$.

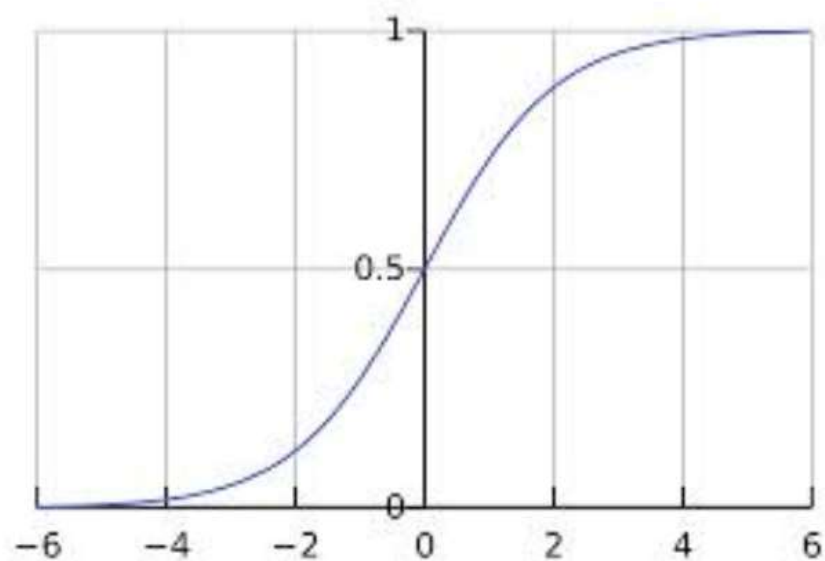
The probability of $y = 1$:

- Increases as the linear function grows.
- Is 50% when this linear function is zero.

How can we convert $w \cdot x + b$ into a probability?

The squashing function

$$s(z) = \frac{1}{1 + e^{-z}}$$



Logistic regression model

Binary labels $y \in \{-1, 1\}$. Model:

$$\Pr(y = 1|x) = \frac{1}{1 + e^{-(w \cdot x + b)}}$$

What is $\Pr(y = -1|x)$?

Logistic regression for binary labels

- Data $x \in \mathbb{R}^d$
- Binary labels $y \in \{-1, 1\}$

Model parametrized by $w \in \mathbb{R}^d$ and $b \in \mathbb{R}$:

$$\Pr_{w,b}(y|x) = \frac{1}{1 + e^{-y(w \cdot x + b)}}$$

Learn parameters w, b from data

The learning problem

Maximum-likelihood principle: given data $(x^{(1)}, y^{(1)}), \dots, (x^{(n)}, y^{(n)}) \in \mathbb{R}^d \times \{-1, 1\}$, pick $w \in \mathbb{R}^d$ and $b \in \mathbb{R}$ that maximize

$$\prod_{i=1}^n \Pr_{w,b}(y^{(i)} \mid x^{(i)})$$

Take log to get **loss function**

$$L(w, b) = - \sum_{i=1}^n \ln \Pr_{w,b}(y^{(i)} \mid x^{(i)}) = \sum_{i=1}^n \ln(1 + e^{-y^{(i)}(w \cdot x^{(i)} + b)})$$

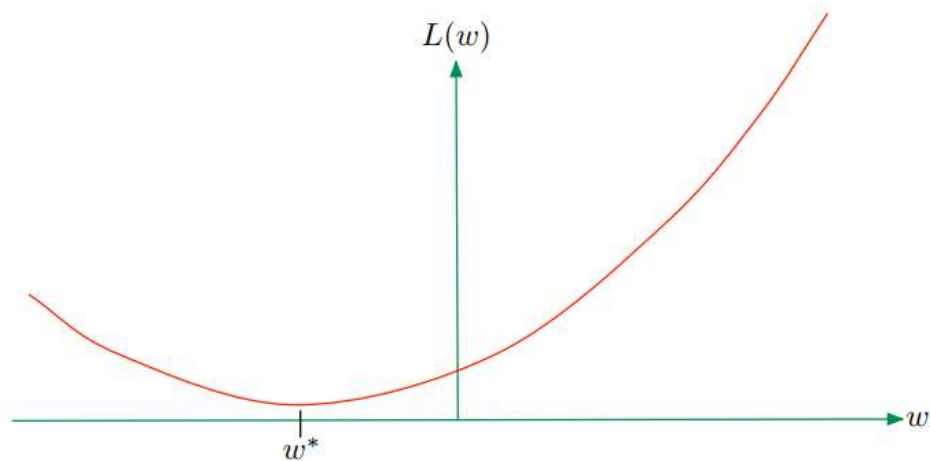
Goal: minimize $L(w, b)$.

As with linear regression, can absorb b into w .

Yields simplified loss function $L(w)$.

Convexity

- Bad news: no closed-form solution for w
- Good news: $L(w)$ is **convex** in w



How to find the minimum of a convex function? By **local search**.

Gradient descent

Given $(x^{(1)}, y^{(1)}), \dots, (x^{(n)}, y^{(n)}) \in \mathbb{R}^d \times \{-1, 1\}$, find

$$\arg \min_{w \in \mathbb{R}^d} L(w) = \sum_{i=1}^n \ln(1 + e^{-y^{(i)}(w \cdot x^{(i)})})$$

- Set $w_0 = 0$
- For $t = 0, 1, 2, \dots$, until convergence:

$$w_{t+1} = w_t + \eta_t \sum_{i=1}^n y^{(i)} x^{(i)} \underbrace{\Pr_{w_t}(-y^{(i)} | x^{(i)})}_{\text{doubt}_t(x^{(i)}, y^{(i)})},$$

where η_t is a “step size”

Use of logistic regression

- Text classification
- Bag-of-words representation for text
- Solution by logistic regression
- Margin versus test error
- Interpreting the model

Sentiment data

Data set: sentences from reviews on Amazon, Yelp, IMDB.
Each labeled as positive or negative.

- Needless to say, I wasted my money.
- He was very impressed when going from the original battery to the extended battery.
- I have to jiggle the plug to get it to line up right to get decent volume.
- Will order from them again!

2500 training sentences, 500 test sentences

Handling text data

Bag-of-words: vectorial representation of text sentences (or documents).

It was the best of times, it was the worst of times, it was the age of wisdom, it was the age of foolishness, it was the epoch of belief, it was the epoch of incredulity, it was the season of Light, it was the season of Darkness, it was the spring of hope, it was the winter of despair, we had everything before us, we had nothing before us, we were all going direct to Heaven, we were all going direct the other way – in short, the period was so far like the present period, that some of its noisiest authorities insisted on its being received, for good or for evil, in the superlative degree of comparison only.



1	despair
2	evil
0	happiness
1	foolishness

- Fix V = some vocabulary.
- Treat each sentence (or document) as a vector of length $|V|$:

$$x = (x_1, x_2, \dots, x_{|V|}),$$

where $x_i = \#$ of times the i th word appears in the sentence.

Logistic regression approach

Code positive as $+1$ and negative as -1 .

$$\Pr_{w,b}(y \mid x) = \frac{1}{1 + e^{-y(w \cdot x + b)}}$$

Given training data $(x^{(1)}, y^{(1)}), \dots, (x^{(n)}, y^{(n)}) \in \mathbb{R}^d \times \{-1, 1\}$, find w, b minimizing

$$L(w, b) = \sum_{i=1}^n \ln(1 + e^{-y^{(i)}(w \cdot x^{(i)} + b)})$$

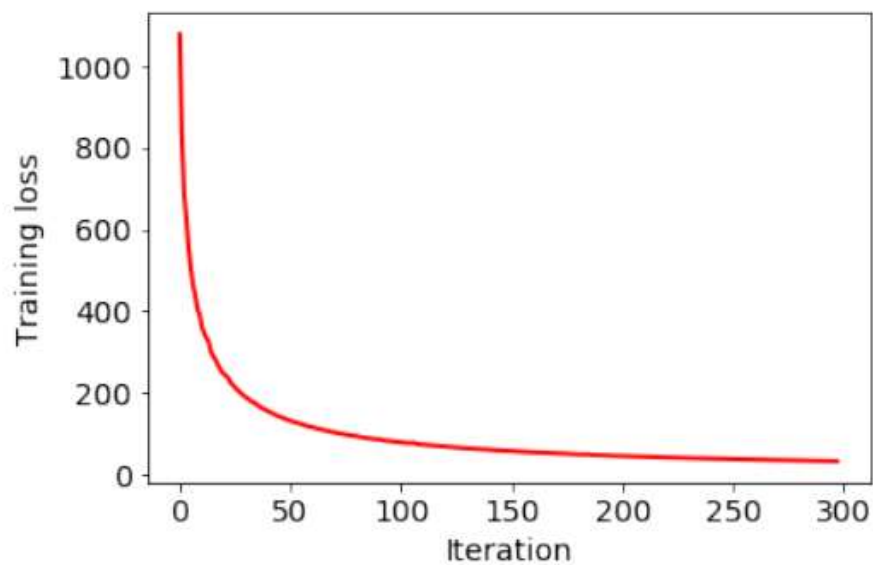
Convex problem with many solution methods, e.g.

- gradient descent, stochastic gradient descent
- Newton-Raphson, quasi-Newton

All converge to the optimal solution.

Local search in progress

Look at how loss function $L(w, b)$ changes over iterations of stochastic gradient descent.



Final model: **test error 0.21.**

Mistakes

Not much dialogue, not much music, the whole film was shot as elaborately and aesthetically like a sculpture. 1

This film highlights the fundamental flaws of the legal process, that it's not about discovering guilt or innocence, but rather, is about who presents better in court. 1

You need two hands to operate the screen. This software interface is decade old and cannot compete with new software designs. -1

The last 15 minutes of movie are also not bad as well. 1

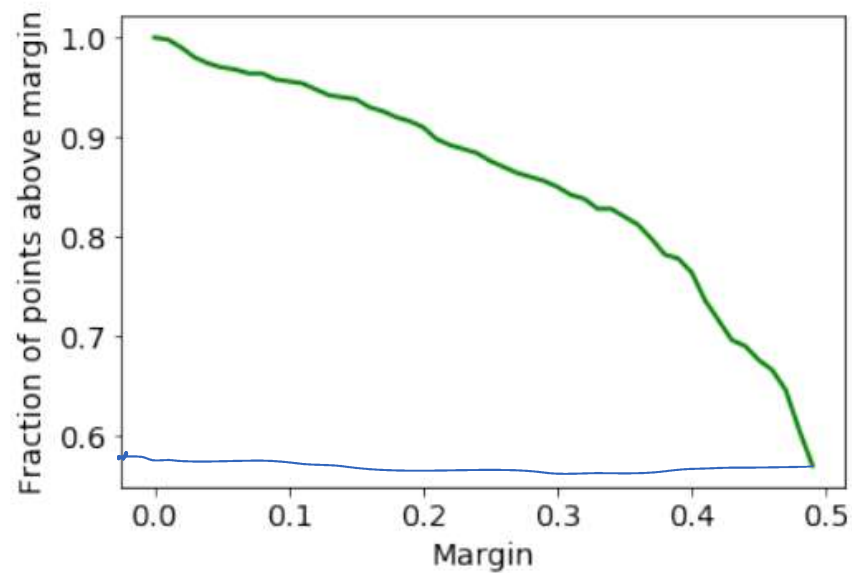
If you plan to use this in a car forget about it. -1

If you look for authentic Thai food, go else where. -1

Waste your money on this game. 1

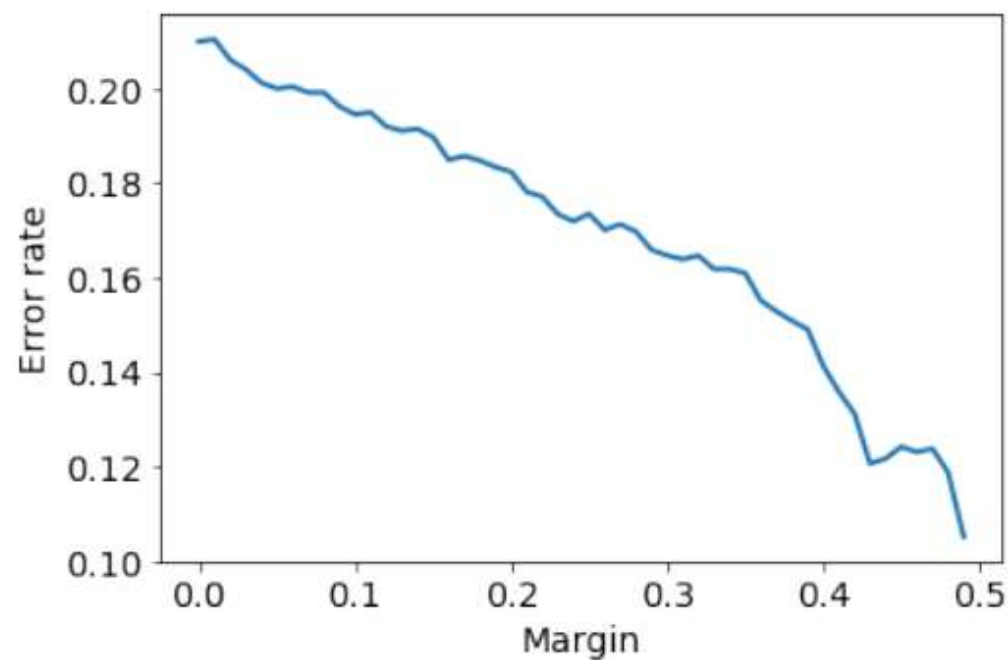
Margin and test error

$$\text{Margin on test pt } x = \left| \Pr_{w,b}(y = 1|x) - \frac{1}{2} \right|$$



Margin and test error

$$\text{Margin on test pt } x = \left| \Pr_{w,b}(y = 1|x) - \frac{1}{2} \right|$$



Interpreting the model

Words with the most positive coefficients

'sturdy', 'able', 'happy', 'disappoint', 'perfectly', 'remarkable', 'animation',
'recommendation', 'best', 'funny', 'restaurant', 'job', 'overly', 'cute', 'good', 'rocks',
'believable', 'brilliant', 'prompt', 'interesting', 'skimp', 'definitely', 'comfortable',
'amazing', 'tasty', 'wonderful', 'excellent', 'pleased', 'beautiful', 'fantastic',
'delicious', 'watch', 'soundtrack', 'predictable', 'nice', 'awesome', 'perfect', 'works',
'loved', 'enjoyed', 'love', 'great', 'happier', 'properly', 'liked', 'fun', 'screamy',
'masculine'

Words with the most negative coefficients

'disappointment', 'sucked', 'poor', 'aren', 'not', 'doesn', 'worst', 'average',
'garbage', 'bit', 'looking', 'avoid', 'roasted', 'broke', 'starter', 'disappointing', 'dont',
'waste', 'figure', 'why', 'sucks', 'slow', 'none', 'directing', 'stupid', 'lazy',
'unrecommended', 'unreliable', 'missing', 'awful', 'mad', 'hours', 'dirty', 'didn',
'probably', 'lame', 'sorry', 'horrible', 'fails', 'unfortunately', 'barking', 'bad', 'return',
'issues', 'rating', 'started', 'then', 'nothing', 'fair', 'pay'